MAHATMA GANDHI UNIVERSITY, KOTTAYAM

BBA (HONOURS)

FIRST SEMESTER EXAMINATION (2024 ADMISION ONWARDS)

MG1CCRBBA102 – BUSINESS STATISTICS AND LOGIC

Duration: 2 hrs **Maximum Marks: 70**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any 5 out of 8 Questions. Each question carries 1 mark.

1. Give one situation in which the geometric mean is more appropriate than the arithmetic mean. [U] [CO1]
2. Select the number that will come next in the following series
    a. 6,9,13,18,24,31, ? [A] [CO2]
3. Define Arithmetic progression. [K] [CO2]
4. A marketing team wants to test if a new online advertisement will increase the likelihood of customers clicking on it. Historically, only 5% of visitors to their site click on ads. The team plans to show the new ad to 20 randomly selected visitors. What type of distribution can be used to determine the ad's effectiveness by checking if the number of clicks on the new ad is significantly higher than what they would expect from the historical rate? [A] [CO3]
5. Identify the central tendency measure best used for:
    a. Summarising monthly purchase data.
    b. Completion times of a series of tasks in a project, where some tasks took significantly longer than others. [U] [CO1]
6. What does a positively skewed customer rating data suggest? [U] [CO1]
7. Find the range of the given values: 5,12,7,4,22,16 [A] [CO3]
8. What would be your interpretation if the correlation coefficient 'r' is such that a) r = -1 b) r = 0.45 [U] [CO5]

Part B

Short Essay Type Questions

Answer any 4 out of 6 Questions. Each question carries 5 marks.

9. What is classification? Enumerate the different methods of classification of data.[K] [CO1]
10. There are 4 men and 3 women. Find the probability of selecting 3 of which
    a. Exactly two are women

b. No woman
c. Atleast one woman
d. Atleast 2 women
e. Atmost 2 women [A] [CO4]

11. A company is analyzing the annual salaries of its employees to decide on appropriate pay adjustments. The salaries range from entry-level positions earning close to minimum wage to executive roles with significantly higher earnings. The HR team wants a measure of "average" salary that best represents a typical employee's earnings.
    a. Which measure of central tendency—mean or median—would be most appropriate for this analysis?
    b. Justify your choice, explaining how the chosen measure would impact decision-making in this context. [A], [CO4]
12. What is Standard Deviation? Discuss the importance of Standard Deviation. [K], [CO1]
13. Distinguish between Geometric mean and Harmonic mean with suitable examples [U], [CO1]
14. Calculate mean deviation about mean from the following data
    x : 10 11 12 13 14

    f : 3 12 18 12 3
    [A], [CO3]

Part C

Essay Type Questionsqw

Answer any 2 out of 4 Questions. Each question carries 20 marks.

15. Explain the differences between correlation and regression with suitable examples. Describe the various types of regression and their distinct applications. [U], [CO5]
16. Create a frequency distribution table based on a set of data. Describe the steps involved in constructing this table, including how to categorize the data and determine the frequency of each category. [A], [CO2]
17. a) Find Karl Pearson's coefficient of correlation between imports and exports from the following data:

Imports	48	68	72	75	80	70	93	100
Exports	64	50	39	48	52	46	40	30

    b) Interpret the result. [An] [CO5]
18. If the insurance claims for automobile accidents follow the given distribution, find Mode:

    Annual Claim (Rs): 0-250, 250-500, 500-750, 750-1000, 1000-1500

    Frequency: 52, 337, 1066, 1776, 1492

[An] [CO4]